Quantum Optics, IPT5340: Homework #1

Ray-Kuang Lee[1]

[1] *Room 911, Delta Hall, National Tsing Hua University, Hsinchu, Taiwan.*
*Tel: +886-3-5742439; E-mail: rklee@ee.nthu.edu.tw**

(Dated: Spring, 2026)

Deadline: 11:59PM, Monday, April 13th, 2026.

Coherent laser fields described by the quantum state $|\alpha\rangle$ can be put into a mesoscopic superposition of two such states,

$$|\text{Cat}\rangle \equiv |\alpha\rangle - |-\alpha\rangle. \tag{1}$$

Here, we only focus on Odd Cats, with the *minus sign* in between. Now, the light field is superposed in such a way that the relative phase between the individual components leads to destructive interference at the center. Such superposition states are called Optical Schrödinger "Cat states, $|\text{Cat}\rangle$" in honor of Erwin Schrödinger's famous *gedanken experiments* to illustrate paradoxa of quantum physics.

Your Task is to create Optical Cat state in Wigner phase space and demonstrate its evolution in time, most likely via **QuTip**.

For **QuTip**, you may refer to P.D. Nation and J.R. Johansson, "QuTiP: The Quantum Toolbox in Python," Release 2.2.0, March 01, 2013.

There are at least two ways to create Optical Cat states:

- Method [1]: Applying Displacement Operator:
  1. Applying Displacement Operator on Vacuum states with $\hat{D}(\alpha)$ and $\hat{D}(-\alpha)$, i.e.,

$$\hat{D}(\alpha)\,|0\rangle = |\alpha\rangle \qquad \text{and} \qquad \hat{D}(-\alpha)\,|0\rangle = |-\alpha> . \tag{2}$$

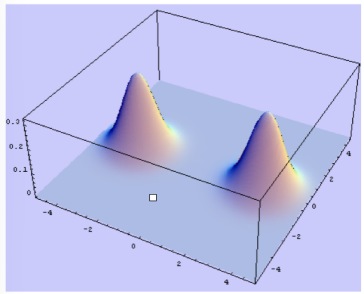

Figure 1: An optical Cat state without superposition.

2. Figure out how to create the *superposition* between $|\alpha>$ and $|-\alpha>$, in order to have the interference in between, i.e.,

$$|\alpha\rangle - |-\alpha\rangle. \tag{3}$$

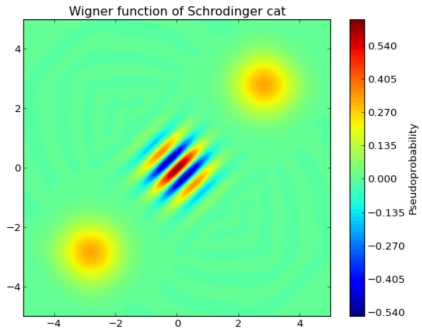

Figure 2: An optical Cat state with superposition.

- Method [2]: Photon-added (or Photon-Subtracted) Squeezed States (here, we only focus on photon addition):
  1. Applying Photon-addition (Creation) Operator on Squeezed Vacuum states, i.e.,

$$|\text{Kitten}\rangle \equiv \hat{a}^\dagger\,|\xi\rangle. \tag{4}$$

Of course, here, you shall have know how to create Squeezed Vacuum states by applying Squeezing Operator $\hat{S}(\xi)$ on Vacuum states, i..e,

$$|\xi\rangle = \hat{S}(\xi)\,|0\rangle. \tag{5}$$

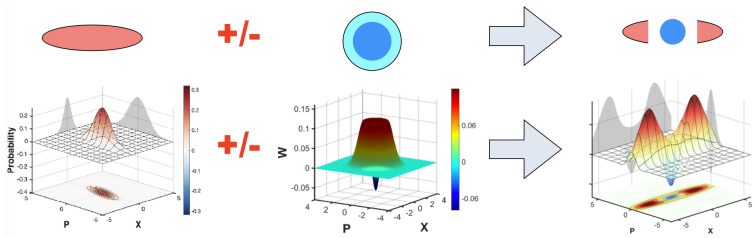

Figure 3: An Optical Kitten state from photon-addition (or subtraction) on squeezed vacuum state.

Your Tasks:

1. Generate Figure 1 for a optical Cat state without superposition.
2. Generate Figure 2 for a optical Cat state with superposition, i.e., $|\text{Cat}\rangle \equiv |\alpha\rangle - |-\alpha\rangle$.
3. Generate Figure 3 for a optical Kitten state by adding one-photon state on squeezed vacuum state, i.e., $|\text{Kitten}\rangle \equiv \hat{a}^\dagger\,|\xi\rangle$.
4. Compare this Kitten state to a Cat state by calculating the *Fidelity*, i.e.,

$$|\,\text{Tr}(\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\,)\,|^2, \tag{6}$$

   here Tr stands for *trace*, $\sigma = |\text{Cat}\rangle\langle\text{Cat}|$, and $\rho = |\text{Kitten}\rangle\langle\text{Kitten}|$.
5. Show the Time-Evolutions of your Cat state.
6. Show the Time-Evolutions of your Kitten state.
7. What are the similarities and differences between your Cat and Kitten states?

*Electronic address: `rklee@ee.nthu.edu.tw`